

# Evaluating User Perception Toward Pose-Changed Avatar in Remote Dissimilar Spaces

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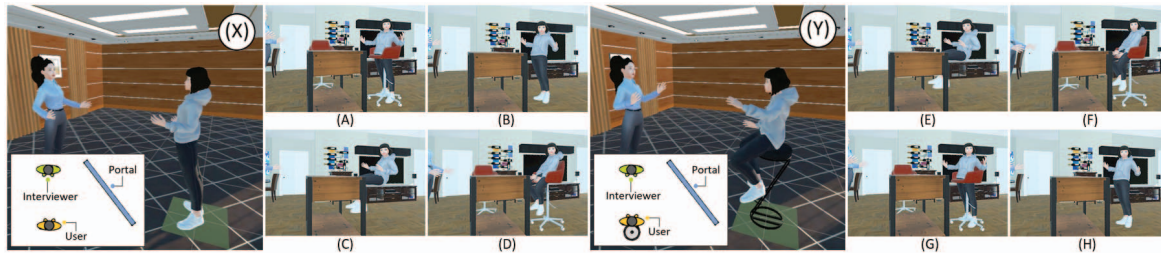


Figure 1: (A)~(H) are the user's 1PP view of the portal of 8 study conditions of our study. (X) and (Y) are 2 levels of the factor Current Pose displayed with 3PP and top view drawing. Note: these images might not be temporally aligned.

## ABSTRACT

In dissimilar telepresence environments, where spaces differ in size and furniture layouts, researchers have proposed methods to adjust avatar movement to deviate from the user's while still adapting to the physical environment. From this perspective, this poster aims to investigate users' perceptions and preferences toward their adjusted avatars. Especially, we propose an experiment focusing on pose standing and sitting which is the most common pose when talking to others. We describe the system setup and initial results that show how users perceive their avatars when their pose category is changed to avoid physics conflict.

**Index Terms:** Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI; Human-centered computing—Human computer interaction (HCI)—Collaborative interaction;

## 1 INTRODUCTION

Avatar-mediated extended reality (XR) telepresence enables distant users to communicate as if they were in the same physical space. Early studies on telepresence did not consider the difference between physical spaces, either by focusing on scenarios where the spatial difference is irrelevant or by assuming identical furniture layouts. Researchers have proposed different methods to overcome this problem, such as finding mutually empty spaces for interaction or placing the avatar based on the user's context. Some approaches adjust the avatar's motion to enable natural interaction with the physical environment or to convey user's context clearly [1, 4, 5], resulting in discrepancies between the user and avatar movement. While improving interaction quality is important, we also consider the need for avatars to stay close to users without causing uncanny distortions.

Specifically, this study focuses on the influence of modifying the

pose category (e.g., sitting and standing) and physics conflict (e.g., penetrating into a physical object) on the user's perception of one's own avatar. The user in the local space was either standing or sitting while interacting with the interviewer. However, the remote space, where the interviewer is, might or might not have a chair in the position where the avatar should be placed, making it necessary to determine whether to preserve the user's pose category or not. Dealing with mutual telepresence, we assume that two users, separated in two spaces, send their avatars to remote space. For clarity, we will focus only on "the local space" with one user (referred to as "the user") and the user's avatar (referred to as "the avatar") located in a "remote space". The remote space has a collaborating partner user who has sent their avatar to the local space. We refer to that collaborating partner as "the interviewer".

## 2 POSE CATEGORY AND PHYSICS CONFLICT

The purpose of our study is to see the effect of pose category change when the avatar faces a physics conflict in remote space to the user. We refer to the participants as "users", except for sections 2.3.

### 2.1 Study Design

We examined the effects of three factors (two levels each) on user perception and preference: Factor 1: User's Current Pose Category (Current Pose), Factor 2: User's Pose Category Preservation (Pose Preservation), and Factor 3: Physics Conflict Existence (Physics Conflict). We chose the two most common pose categories when communicating with others: standing and sitting. For the Current Pose factor, participants either **Stand** or **Sit** while performing the task. For the Pose Preservation factor, we chose **Preserved (P)** and **Not Preserved (NP)**. In **P**, the user's full body tracking, including the pose category, was identically transferred to the avatar. In **NP**, we disabled lower-body tracking and instead forced the lower-body pose to either **Stand** or **Sit** by fixing leg joints' angles. Third, for the Conflict Existence factor, we chose **Exists (E)** and **Doesn't Exist (DE)**. In **Exists (E)**, we created an environment that is physically impossible in reality. For the **Stand** case, we created a visual collision between the avatar and a chair, such that the avatar penetrated the chair. For the **Sit** case, the avatar appeared as if it was floating above the ground without a chair. For **Doesn't Exist (DE)**, we designed an environment that does not violate the law of physics. For the **Stand** case, the avatar was properly standing on the ground without any chair, while for the **Sit** case, the avatar was properly sitting on the

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chair. These consist of 2 (Stand, Sit) x 2 (P, NP) x 2 (E, DE) factorial within-subject design. An actor played the role of the interviewer in our experiment to provide consistent behavioral cues to the user. The first-person perspective (1PP) and third-person perspective (3PP) of each condition in the user experiment are shown in Figure 1. 1PP views are slightly corrected to fix skewness for better visibility.

## 2.2 Study Setup

We used the Unity3D/SteamVR for system development and HTC Vive Pro devices (HMD, 2 controllers, 3 Vive trackers) for tracking and displaying the VR environment for the user experiment. We used characters both for the user and the interviewer from Ready Player Me [2]. We constructed two spaces, one for the interviewer and one for the user. During the task, as in Section 2.3, a user opens the portal to see how their avatar looks in the remote space. Fig. 1 (A) (H) depicts the view inside the portal that the user sees in each condition.

**Local User & Interviewer** We applied inverse kinematics (IK [3]) to data from trackers (1 from head, 2 from controllers and 3 from Vive trackers attached to the pelvis and feet) worn by the user to enable full-body tracking.

**Remote Avatar** Some conditions make the remote avatar take a different pose category than the local user. For conditions that required a preserved pose (A, B, E, F), the user's tracked data (HMD, 2 controllers, 3 Vive trackers) was directly transferred to the remote avatar. For conditions that didn't preserve the user's pose (C, D, G, H), the upper and lower body were generated differently. For upper body tracking, we used the information tracked from the HMD and controllers to apply IK. The lower body poses were fixed to maintain either the standing or sitting pose.

## 2.3 Procedure and Task

Once participants finished wearing an HMD, controllers, and trackers, the experiment began with the **Intro** scene. The purpose of the **Intro** scene was to familiarize participants with the scene settings, including the existence of the remote space and key definitions such as the user and the avatar. Participants first calibrated themselves. Then they moved their arms and legs and also moved around the space to be accustomed to full-body tracking. When the participants felt comfortable, the interviewer initiated a conversation by asking the participants to define "research/job" for themselves. After the participant answered, we guided them to open the portal and explained how they can observe their avatar in the remote space using the portal.

Participants experienced 9 scenes in total (Intro + 8 conditions). We randomized the order of conditions for every participant. For the **Stand** cases, participants stood during the experiment. For the **Sit** cases, participants sat on a tall chair in the real environment, as they were sitting on a virtual tall chair in the VR environment. This was to help them imagine that they were physically located inside of the space. The task was questioning and answering between the interviewer and the participants. Each scene began with the interviewer asking two questions and the participants' answers. During this period, participants didn't know how their avatar looked since the portal was not visible. It was only after when participants finished answering two questions, the participants opened the portal and saw the avatar in the remote space. Participants observed both the user and avatar for as long as they needed. Last, the interviewer asked a third question and the participants answered. After experiencing each condition, except for the intro, participants filled out a questionnaire using a 7-point Likert scale. The total experiment procedure took approximately 90 minutes per participant.

We measured 6 components using questionnaires adopted from previous studies: Sense of Ownership (SoO), Sense of Agency (SoA), Trust, Intention Preservation (Intention), Physicality, Preference. We used a 7-point Likert scale. We recruited 25 participants

(10 male, 15 female; M=23.3, SD=2.39).

## 3 INITIAL RESULTS

As the data were not normally distributed according to a Shapiro-Wilk test, we used the nonparametric Wilcoxon signed-rank tests to analyze the effect of each factor. Current Pose did not show significance on any measure, while **Pose Preservation**, and **Physics Conflict** showed significant differences between levels.

**Pose Preservation** Overall, participants' ratings were higher when their pose was preserved (P) compared to when it was not preserved (NP). First, measures related to avatar control and ownership were higher in P than in NP. Sense of ownership (SoO) was higher in P ( $\mu = 4.37$ ,  $\sigma = 0.90$ ) than in NP ( $\mu = 3.75$ ,  $\sigma = 0.90$ ) [ $p < 0.001$ ]. Sense of agency (SoA) also increased in P ( $\mu = 4.10$ ,  $\sigma = 0.60$ ) than NP ( $\mu = 3.61$ ,  $\sigma = 0.67$ ) [ $p < 0.001$ ]. Second, measures related to the system were higher in P. Trust increased in P ( $\mu = 4.42$ ,  $\sigma = 0.80$ ) compared to NP ( $\mu = 3.66$ ,  $\sigma = 1.03$ ) [ $p < 0.001$ ], as well as intention with a score of ( $\mu = 5.42$ ,  $\sigma = 1.11$ ) in P than in NP ( $\mu = 3.90$ ,  $\sigma = 1.48$ ) [ $p < 0.001$ ]. Physicality was higher in P ( $\mu = 4.37$ ,  $\sigma = 0.86$ ) compared to NP ( $\mu = 3.91$ ,  $\sigma = 0.90$ ) [ $p < 0.001$ ]. Last, participants reported a higher preference for cases when their pose was preserved ( $\mu = 4.74$ ,  $\sigma = 1.50$ ) compared to when it was not preserved ( $\mu = 3.38$ ,  $\sigma = 1.55$ ) [ $p < 0.001$ ].

**Physics Conflict** In general, participants reported higher values when there were no physics conflicts (DE) compared to cases where physics conflict existed (E). SoO was higher in DE ( $\mu = 4.21$ ,  $\sigma = 0.94$ ) compared to E ( $\mu = 3.91$ ,  $\sigma = 0.95$ ) [ $p < 0.001$ ]. Second, measures related to the system were higher in DE. Trust increased in DE ( $\mu = 4.29$ ,  $\sigma = 0.97$ ), compared to the E ( $\mu = 3.80$ ,  $\sigma = 0.98$ ) [ $p < 0.001$ ]. Intention preservation perception was also high in DE ( $\mu = 4.87$ ,  $\sigma = 1.54$ ), compared to E ( $\mu = 4.46$ ,  $\sigma = 1.47$ ) [ $p < 0.002$ ]. Physicality increased in DE ( $\mu = 4.39$ ,  $\sigma = 0.91$ ), compared to the E ( $\mu = 3.90$ ,  $\sigma = 0.84$ ) [ $p < 0.001$ ]. Lastly, participants reported higher preference in DE ( $\mu = 4.69$ ,  $\sigma = 1.55$ ), compared to E ( $\mu = 3.43$ ,  $\sigma = 1.55$ ) [ $p < 0.001$ ].

## 4 CONCLUSION

In this poster, we explored user preference and perceptions toward the avatar where synchronization between a user and an avatar causes the avatar to violate the laws of physics in remote space. Users reported feelings of less ownership and control over their avatar, a decreased level of intention and trust toward the system, and less preference when their pose category was modified. Physics conflict also hurts users' perceptions.

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## REFERENCES

- [1] T. Piumsomboon, G. A. Lee, J. D. Hart, B. Ens, R. W. Lindeman, B. H. Thomas, and M. Billingham. Mini-me: An adaptive avatar for mixed reality remote collaboration. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, CHI '18, p. 1–13. Association for Computing Machinery, New York, NY, USA, 2018. doi: 10.1145/3173574.3173620
- [2] ReadyPlayerMe. Metaverse 3d avatar creator.
- [3] RootMotion. Final ik - rootmotion, 2017.
- [4] A. Ullal, A. Watkins, and N. Sarkar. A multi-objective optimization framework for redirecting pointing gestures in remote-local mixed/augmented reality. In *Proceedings of the 2022 ACM Symposium on Spatial User Interaction*, SUI '22. Association for Computing Machinery, New York, NY, USA, 2022. doi: 10.1145/3565970.3567681
- [5] L. Yoon, D. Yang, C. Chung, and S.-H. Lee. A full body avatar-based telepresence system for dissimilar spaces. *ArXiv*, abs/2103.04380, 2021.